

VATHSARAN YASOTHARAN

Data Scientist & Machine Learning Engineer

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As a Data Scientist, I transform complex datasets into production-ready models that drive measurable business impact. **£5.81M revenue intelligence | 90%+ model accuracy | 541k+ records processed.**
Skilled in customer analytics, predictive modeling, and stakeholder communication. Available immediately for data science, ML engineering, or analytics roles.

CORE COMPETENCIES

Machine Learning:

Scikit-learn, XGBoost, Random Forest, SVM, Ensemble Methods, Feature Engineering, RFM Analysis

Programming:

Python, SQL, Pandas, NumPy

Data Engineering:

Data Cleaning, Imputation, Outlier Detection, PCA, Feature Selection (RFE)

Statistics:

Hypothesis Testing, A/B Testing, Regression Analysis, AUC-ROC, Cross-Validation

Communication:

Technical Presentation, Stakeholder Reporting, Data Storytelling, Cross-functional Collaboration

Leadership:

Team Coordination, Project Management, Mentoring, Strategic Planning

Visualization:

Streamlit, Plotly, Seaborn, Matplotlib

Tools:

Git, Docker, VS Code, Jupyter, Google Colab

DATA SCIENCE PROJECTS

E-commerce Revenue Intelligence Platform

Advanced Customer Segmentation & Analytics | Python, Scikit-learn, Streamlit

- Engineered RFM segmentation model analyzing **541k transactions**, identifying **831 VIP customers (19%)** generating **£5.81M (69.5% of total revenue)**
- Achieved **97.8% data quality** through advanced cleaning pipelines; uncovered Q4 seasonality driving **37.8% annual revenue** to optimize inventory strategy
- Deployed interactive Streamlit dashboard enabling real-time customer segmentation with Plotly visualizations
- **Technologies:** Python, Scikit-learn, Pandas, Streamlit, Plotly

Clinical Heart Disease Prediction System

Machine Learning Classification | Scikit-learn, Random Forest, Healthcare Analytics

- Built Random Forest classifier achieving **90.16% accuracy** and **0.951 AUC** on Cleveland Clinic dataset for early heart disease detection

- ▶ Reduced feature space from 13 to 8 using Recursive Feature Elimination (RFE) while maintaining model performance
- ▶ Demonstrated binary classification superiority over multi-class through hyperparameter optimization and learning curve analysis
- ▶ **Technologies:** Scikit-learn, Random Forest, IQR Outlier Capping, RFE, Cross-Validation

Autonomous Vehicle Public Perception Study

Mixed-Methods Research & Statistical Modeling | Python, statsmodels, Regression Analysis

- ▶ Conducted mixed-methods analysis with **100% hypothesis validation ($p < 0.001$)**; regression models explained **91.9% variance** in safety perception
- ▶ Identified multi-modal sensor fusion as primary trust driver through quantitative modeling and qualitative theme integration
- ▶ Delivered comprehensive stakeholder report combining statistical rigor with actionable insights
- ▶ **Technologies:** Python, statsmodels, Regression Analysis, Cohen's d, Thematic Coding

Drone Fleet Predictive Analytics Dashboard

Time-Series Forecasting & Business Intelligence | Streamlit, Prophet, Plotly

- ▶ Developed 4-page Streamlit analytics platform with Prophet time-series forecasting for sales and maintenance optimization
- ▶ Analyzed 1k-record dataset revealing **Africa as top market (23% share)** and **May peak (52k units)**; achieved 100% data completeness via advanced imputation
- ▶ Optimized dashboard for **<3s load time** with responsive Plotly visualizations
- ▶ **Technologies:** Pandas, NumPy, Prophet, Streamlit, Plotly

Titanic Survival Ensemble Model

Advanced Ensemble Learning & Feature Engineering | Scikit-learn, VIF Analysis

- ▶ Implemented soft-voting ensemble (Random Forest, SVM, KNN) achieving **83.8% cross-validation accuracy**
- ▶ Engineered features including Title Group and Has Cabin; managed multicollinearity with VIF analysis and correlation heatmaps
- ▶ Predicted **36.12% survival rate** on unseen test data with robust validation pipeline
- ▶ **Technologies:** Scikit-learn, Feature Engineering, Ensemble Methods, VIF Analysis

LEADERSHIP & COMMUNICATION EXPERIENCE

Sub-Leader & Coordinator – IoT Professional Training Program

Esoft Metro Campus | Colombo, Sri Lanka

- ▶ Co-led **19-member cross-functional team** in designing and delivering full-day IoT workshop, serving as primary coordinator ensuring seamless task delegation and timeline adherence across **2-month project lifecycle**
- ▶ Translated complex technical concepts into accessible presentations for non-technical audience, demonstrating ability to communicate data insights to diverse stakeholder groups
- ▶ Orchestrated multi-channel communication strategy utilizing virtual meetings, collaborative documents (SharePoint), and agile methodologies, resulting in **100% on-time deliverable completion**

- Presented technical findings to senior leadership, receiving commendation for clarity in explaining IoT applications, sensor networks, and data analytics concepts
- **Technologies:** Project Management, Stakeholder Communication, Technical Documentation, Presentation Design

EDUCATION & CERTIFICATIONS

Higher National Diploma in Data Analytics

ESOFT University, Colombo | Pearson-assured (UK Standard)

Diploma in Information Technology (DICT)

ESOFT University, Colombo

Edexcel IGCSE O-Levels

International Qualifications | Physics, Biology, Chemistry, Mathematics, English, ICT

Available immediately for data science, machine learning, or analytics roles in Sri Lanka or remote (Europe, Middle East, APAC)